

FOREWORD

When I was approached by Al to write the foreword to this book, I was pretty excited about the prospect. A book on recursion! Now, that's something you just don't see every day. Considered by many to be one of the more mysterious topics in programming, recursion is often discouraged. Oddly, this stands in stark contrast to its storied use in weird job interview questions.

However, there are all sorts of practical reasons to learn about recursion. Recursive thinking is very much a mindset about problem-solving. At its core, larger problems get broken into smaller problems. And sometimes along the way, hard problems are rewritten into equivalent but easier-to-solve simple problems. This sort of thinking can be a useful tool when applied to software design—even when recursion is not being used. Thus, it's a worthy topic of study for programmers of all skill levels.

In my unbridled excitement to say more about recursion, I originally wrote this foreword in the form of a few short stories involving friends who'd applied recursive thinking in different ways but achieved a similar result. First there was the story of Ben, who learned about recursion, took it too far, and somehow managed to disappear off the face of the earth under mysterious circumstances after committing the following Python code into production:

```
result = [(lambda r: lambda n: 1 if n < 2 else r(r)(n-1) + r(r)(n-2))(  
    (lambda r: lambda n: 1 if n < 2 else r(r)(n-1) + r(r)(n-2)))(n)  
    for n in range(37)]
```

Then there was the story of Chelsea, who became so effective at real-world problem-solving that she was promptly fired! Oh, you wouldn't believe how much all the fine editors at No Starch (bless their hearts) hated these stories. "You can't start a book by telling people stories like that. It's